

DATA - THE NEW OIL - Transforming Vessel Performance Through AI and Machine Learning



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Abstract:

This article examines the transformative role of **data analytics and artificial intelligence/machine learning (AI-ML)** in optimising vessel operations within the maritime industry. Traditional fuel consumption estimation methods rely heavily on manual compilation of speed-fuel matrices derived from logbooks and noon reports, a process that is labour-intensive and often limited in accuracy. The paper proposes a transition toward **data-driven decision-making frameworks** that utilise high-frequency sensor data and structured operational datasets generated onboard modern vessels.

Using machine learning techniques such as **multiple linear regression, decision trees and clustering algorithms**, predictive models can estimate fuel consumption with significantly improved accuracy by incorporating operational variables including vessel draft, engine load, weather conditions and speed parameters. The study also evaluates the comparative advantages of **high-frequency sensor data versus conventional low-frequency noon report data**, while addressing key challenges related to data quality, sensor calibration and data governance.

A central theme of the article is the emerging role of **seafarers transitioning ashore**, who can leverage maritime domain expertise alongside data science skills to develop effective predictive analytics systems. The study concludes that structured maritime datasets analysed through AI-ML can substantially improve operational

efficiency, reduce fuel costs, support decarbonisation initiatives and accelerate the industry's progress toward **net-zero emissions**, positioning data as a strategic resource in the future of maritime operations.

Keywords: Data analytics, artificial intelligence, machine learning, fuel consumption optimisation, vessel operations, predictive modeling, maritime industry, voyage optimisation, high-frequency data, noon reports, operational efficiency, decarbonisation, emissions reduction, decision trees, regression analysis, sensor data, performance benchmarking, hull fouling, maritime domain expertise, net-zero emissions.

Introduction:

Predicting future events based on historical data has always been a subject of great interest. By analysing past trends and patterns, we gain valuable insights into how things may unfold in the future.

While there's no guarantee that history will repeat itself, we still rely on these projections to make informed business decisions, hoping that circumstances will align favorably.

Uncertainties are an inevitable part of this process, yet they, too, contribute to the historical data that will shape tomorrow's forecasts.

The marine industry, in particular, faces numerous unpredictable factors that significantly impact business decisions.

Weather conditions, geopolitical developments, port activities, regulatory changes, accidents, market fluctuations and more all play a role, none of which can be precisely timed or predicted.

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Data transforms ship operations into predictive intelligence
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Vessel operations are heavily influenced by commercial considerations, with a constant focus on optimising fuel consumption and speed to enhance profitability and reduce costs.

In this context, “What-if” scenarios serve as crucial tools for evaluating key variables.

Current Approach- Random sampling:

Those familiar with the process know the painstaking effort required to populate speed-fuel consumption matrices repeatedly.

These matrices help estimate fuel usage under various conditions—ballast or laden voyages, tank cleaning, cargo heating, inerting, purging, port operations, anchorage stays and more.

To derive the most accurate estimates, old logbooks, log abstracts and noon reports must be meticulously reviewed.

The resulting data is then shared with owners, charterers, managers and operators, often forming the basis for charter party agreements and operational contracts.

More often than not, the estimates prove unreliable—sometimes veering so far from actual performance that the exercise devolves into a copy-paste routine with little operational relevance.



Illustration: Cumbersome manual estimations

Data-Driven Decision making with AI-ML:

With the wealth of historical data available today and advancements in “AI and machine learning (ML)”, we can now generate far more accurate estimates and actionable insights.

Instead of relying solely on manual data compilation, we can leverage high-frequency sensor data or structured noon reports to run dynamic “What-if” scenarios. Key variables such as:

- Draft
- Engine load
- Depth under keel
- Weather conditions (wind speed/direction, wave height, swell characteristics)
- Shaft power & RPM
- Speed over ground vs. speed through water
- Current speed & direction

These data points can be analysed to predict fuel consumption based on their correlations.

By applying machine learning models—such as multiple linear regression, decision trees and clustering techniques—we can derive highly reliable estimates that adapt to real-world conditions. These models not only enhance accuracy but also identify hidden patterns that traditional methods might miss, leading to better decision-making in voyage optimisation, cost reduction and emissions control.

This data-driven approach minimises guesswork, improves efficiency and provides a competitive edge in vessel operations. Refer **Figure.1** for ‘What-If’ variables used in analysis.

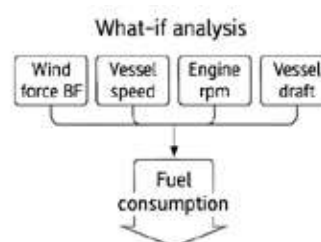


Figure. 1. What-if analysis

Benchmarking Operational Performance:

Operational data can be compared against shop test or sea trial results to assess conformance and identify deviations.

This analysis helps evaluate real-world performance against baseline expectations, enabling timely adjustments and optimising efficiency.

However, it is important to recognise that sea trials and shop tests provide limited reference points. Conducted under controlled, fixed conditions, these results are inherently restrictive.

In contrast, AI-ML models leverage a “living baseline” derived from high-frequency historical data, allowing for benchmarking across the full spectrum of real-world variables such as varying drafts, aging hulls and fluctuating weather patterns, that static sea trials simply cannot account for.

Leveraging Historical Data for Smarter Baseline Modeling:

Applying pattern recognition techniques to historical operational data enables the creation of more accurate and realistic performance baselines.

This data-driven approach moves beyond theoretical assumptions, providing actionable benchmarks that reflect real-world conditions and variability.

Fuel Consumption Optimisation for Vessels using AI-ML:

As shown in **Figure. 2**, by analysing historical data, accurate fuel consumption estimates can be derived for different operating conditions, including speed, draft, RPM and weather. The predictive analytics solution enables precise fuel consumption modeling across all operational parameters - speed, draft, RPM and weather conditions.

This powerful capability provides both shipboard teams and commercial operators with:

- Decision-Ready Intelligence - Forecast fuel needs for any voyage scenario with unprecedented accuracy.
- Operational Transparency - Visualise how each variable impacts consumption in real-world conditions.
- Strategic Advantage - Base critical decisions on data-driven projections rather than estimates.
- Cost Control - Identify and execute the most fuel-efficient operating profiles.
- By transforming complex operational data into actionable insights, the system serves as both a navigation companion for engineers and a business intelligence tool for commercial teams - bridging the gap between ship operations and financial performance.
- The true value lies in its ability to convert theoretical models into practical, voyage-specific guidance that drives measurable efficiency gains across the fleet.

Comparing past and present “what-if” scenarios can also help detect engine wear & tear, hull and propeller degradation due to fouling, fleet wise or sister vessel performance comparisons.

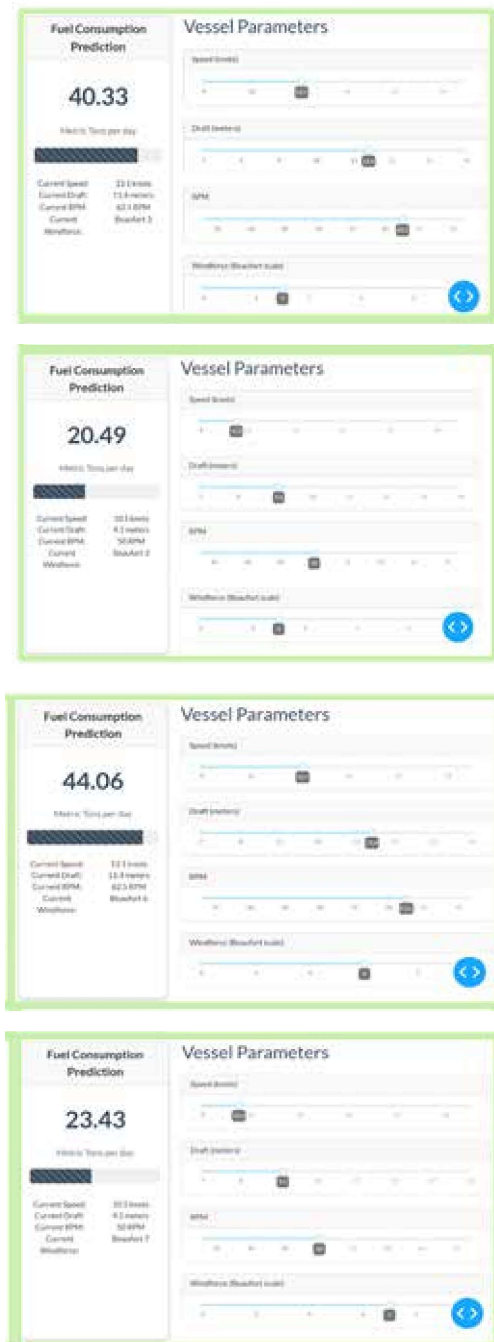


Figure.2 A typical 'What-if application' to predict fuel consumptions using historic data

These insights enable better planning, business decisions, optimised vessel operations and significant cost savings.

Practical Case Study: Detecting the Efficiency Gap

To illustrate the power of AI-ML in real-world operations, consider a vessel following its drydocking. After several initial voyages, the captured historical data once

AI reveals hidden patterns in vessel performance

Parameter	Post-Drydock (Baseline)	Scenario A: Minimum Variance	Scenario B: High Variance
Vessel Speed	12.0 Knots	12.0 Knots	12.0 Knots
Laden Draft	11.5 Meters	11.5 Meters	11.5 Meters
Engine RPM	80 RPM	80 RPM	80 RPM
Weather (BF)	Good (BF 3)	Good (BF 3)	Good (BF 3)
Fuel Consumption	25.0 MT/Day	25.4 MT/Day	26.5 MT/Day
Variance (%)	0.0%	+1.6%	+6.0%
Additional Daily Cost	–	\$240	\$900
Voyage Loss (20 Days)	–	\$4,800	\$18,000

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Table 1. Comparing operating trends over time

meticulously cleaned to remove outliers and noise, is processed through various machine learning algorithms. By ensuring all key parameters are available, we minimise uncertainty and establish a credible performance baseline.

The Baseline Scenario:

- Operating Conditions: 12 knots speed, 11.5m laden draft, 80 engine RPM.
- Environment: Deep sea, Good Weather (Beaufort 3), negligible currents.
- Initial Performance: The model confirms a fuel consumption of **25 MT/day**.

The AI-ML Insight: Using this baseline as training data, the AI continuously evaluates future voyages. If, four months later, the vessel encounters identical operating and weather conditions but consumes **26.5 MT/day**, the system immediately flags a **6% increase** in consumption. Refer table 1. A case of minimum variance is also shown.

The Impact:

- Financial Leakage: This seemingly small deviation translates to an additional **\$18,000** in fuel costs for a single 20-day voyage.
- Proactive Management: Continuous “What-if” analysis allows the Chief Engineer and commercial teams to quantify hull and propeller degradation in real-time.

The Gold Standard for Accuracy: Whether utilising high-frequency (HF) sensor data or structured noon reports (LF), the credibility of these predictions’ hinges on **data quality**.

Ensuring the availability of all critical parameters is the only way to reduce the margin of error and transform raw data into a reliable navigation and business tool.

Conversely, for maintaining the same consumption, a loss of speed could be observed if the focus of analysis and quantification is for loss of speed as shown in Table 2.

High-Frequency Data vs. Noon Reports: Finding the Right Balance:

This topic remains widely debated, with strong arguments on both sides.

The answer isn’t straightforward—it depends on the specific needs and expectations of the user. Refer Table 3.

Rather than dismissing either high-frequency (HF) data or noon reports (low-frequency/LF data), it’s important to assess what insights each can deliver.

Too much information (TMI) can overwhelm users, leading to disengagement.

Some service providers clutter dashboards with excessive data, making it difficult to extract meaningful insights.

Analysis Focus	Parameter	Baseline	Evaluation (4 Months Later)	Resulting Loss
Fuel Focus	Consumption (at 12 knots)	25.0 MT	26.5 MT	+6% Fuel Cost
Speed Focus	Speed (at 25 MT/day)	12.0 Kts	11.4 Kts	-0.6 Kts (Time Loss)

Table 2. Focusing on fuel consumptions & speed of vessel.

Comparison Table: Merits and Demerits:

Feature	High-Frequency (HF) Data	Noon Reports (LF Data)
Data Resolution	Continuous, granular readings (e.g., every minute/hour)	Single snapshot per day (typically at noon)
Accuracy & Timeliness	Real-time insights reduce lag and improve responsiveness	Prone to delays and human error; lacks operational granularity
Operational Visibility	Enables predictive analytics, anomaly detection, and trend analysis	Limited visibility; useful for broad summaries and historical tracking
Infrastructure Requirements	Requires sensors, connectivity, data storage, and analytics platforms	Minimal tech requirements; relies on manual entry
User Engagement	Can overwhelm users if dashboards are cluttered or poorly designed	Easier to digest; familiar format for most operators
Cost & Complexity	Higher upfront investment; ongoing maintenance and training needed	Low-cost and simple to implement
Use Cases	Ideal for performance optimization, emissions tracking, and dynamic routing	Suitable for compliance reporting, charter party documentation, and basic monitoring

Table 3. Feature comparisons



Ultimately, both HF and LF data have value—the choice depends on the client’s objectives and their willingness to invest in the necessary infrastructure.

Challenges and Considerations:

- **Data Quality:** Reliable mathematical models depend on high-quality data.
- **Sensor Accuracy:** Regular calibration of key instruments is essential to ensure precise measurements.
- **Data Acquisition Issues:** Faulty installations, network disruptions and scaling errors can introduce inaccuracies, affecting computations and modeling.
- **Data Cleaning:** Inefficient handling of outliers and noise can undermine the credibility of models.
- **Domain Expertise:** Understanding the relevant variables is crucial for effective modeling.

Elevating the Human Element: Human in the loop

Will AI replace humans? The transition from traditional logbooks to AI-driven operations is often met with the fear of automation replacing human expertise. However, in the maritime context, AI-ML is not a replacement for the **Engineer**; it is an evolution of the engine room team.

Think of AI as a **“Digital Assistant Engineer”** working 24/7 in the background. While the human Engineer manages physical maintenance and daily operations, the AI handles the “digital drudgery”.

- **Monotonous Data Crunching:** Instead of a human spending hour manually reviewing old log abstracts

and noon reports to populate speed-fuel matrices, the AI processes thousands of data points from high-frequency sensors in seconds.

- **Predictive Diagnostics:** The system identifies subtle deviations in machinery health or hull efficiency that might be invisible to the naked eye, acting as a navigation companion for the engineering team.
- **Focus on High-Level Maintenance:** By automating the “What-if” scenarios and consumption forecasts, the AI frees the human Engineer to focus on critical decision-making, complex repairs and ensuring the vessel meets stringent net-zero emission goals.

Upskilling: AI as the Modern “Screwdriver”

Just as a marine engineer must be trained to use a laser alignment tool or a hydraulic jack, AI must be viewed as a ‘technical tool’ that requires a specific skill set to operate effectively.

- **Mechanical vs. Digital Tools:** Working mechanically requires knowledge of physical tolerances and material science; working with AI requires “digital domain expertise”, an understanding of how data translates into physical performance.
- **The Necessity of Upskilling:** For AI to be a truly effective tool, seafarers transitioning ashore must upskill in areas like **statistics, advanced excel, coding (Python, R) & data visualisation (Power BI, Tableau)**.
- **Bridging the Gap:** Relying on data scientists who lack sea-time often leads to ineffective models. **The most powerful results come from professionals who combine maritime domain knowledge with data**

science tools, ensuring the AI's "output" makes sense in a "real-world" engine room.

"As a seafarer and data science practitioner, I've seen firsthand how this dual expertise drives faster, more impactful results."

Navigating Regulatory Waters with AI-ML:

The maritime industry is currently governed by IMO's short-term measures aimed at reducing greenhouse gas emissions.

While **EEXI** is a technical "one-time" certification of a ship's design, **CII** is an ongoing operational rating that measures how efficiently a ship transports goods.

- **The CII Challenge:** A vessel is rated from "A" to "E" based on its annual carbon intensity. A ship that performs well today could slip into a "D" or "E" rating next year due to hull fouling, engine degradation, or poor voyage planning.
- **AI as a Rating Shield:** This is where data-driven "What-if" scenarios become critical. Instead of waiting for the end of the year to discover a poor rating, AI models use high-frequency or structured noon data to provide a **Real-Time CII Forecast**.
- **Precision Decarbonisation:** By analysing variables like speed, draft and weather, the ML model identifies the "Sweet Spot" the exact operating profile where fuel consumption is minimised without compromising the commercial schedule.
- **Impact on Charter Value:** A vessel maintained in the "A" or "B" category via AI-optimised operations commands higher charter rates and remains compliant with environmental standards, potentially unlocking financial benefits like carbon credit trading. Ships operating efficiently can lower their carbon allowances,

Analytics powers the next revolution in shipping efficiency

reduce costs or even profit from trading excess allowances. Compliance with stricter EU fuel standards reduces penalties and enhances market competitiveness.

How AI Directly Improves the CII Rating:

The formula for CII is essentially the ratio of CO₂ emitted to the "work" done in transportation (capacity x distance). Since CO₂ emissions are directly proportional to fuel consumption, AI improves the rating by:

1. **Optimising the "What-if" of Voyage Planning:** Calculating the most fuel-efficient route and speed based on predicted weather and currents to minimise the "Numerator" (Total Fuel Burned).
2. **Early Fouling Detection:** As seen in our case study, a 6% increase in fuel due to hull fouling doesn't just cost \$18,000; it can push a vessel from a "C" to a "D" rating. AI detects this trend early, allowing for a mid-cycle hull cleaning that restores the rating.
3. **Engine Performance Benchmarking:** Comparing real-world data against shop tests ensures the engine is operating at peak thermal efficiency, reducing unnecessary emissions.

For a commercial operator, data is no longer just for the technical department. **It is the difference between a vessel being a "prime asset" or a "stranded asset" in a net-zero world.**

Conclusion:

A well-maintained operational database analysed using **AI and machine learning** effectively becomes the *digital twin* of a vessel's physical performance. This approach moves maritime operations beyond the limitations of manual noon reports toward a system of **continuous, data-driven insight**. By identifying patterns in fuel consumption and predicting issues related to machinery health, hull fouling and propeller efficiency, AI-driven analytics converts raw data into actionable operational intelligence.

The benefits of this transformation are twofold.

1. **Financial resilience:** Predictive analytics minimises the "efficiency gap" caused by unnoticed degradation and suboptimal operating conditions, enabling operators to reduce fuel costs and improve voyage profitability.
2. **Environmental leadership:** Optimised fuel consumption directly reduces carbon emissions, helping vessels maintain strong **CII ratings** and comply with **EEXI and future decarbonisation regulations**. As regulatory frameworks tighten, efficient vessels may



also gain access to **carbon credit markets and green financing opportunities**.

This transformation is creating a new professional role—the **Maritime Data Practitioner**—who combines maritime domain expertise with data science skills to translate operational data into strategic insights.

If data is the new oil, then analytics is the new combustion engine. Those who master this capability will not only improve operational efficiency but also guide the maritime industry toward a more sustainable and data-driven future.

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About the Author



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